

ABOUT ME

Self Introduction Video

Results-driven computer engineer with a passion for building performant software and leading research projects from idea to deployment. Passionate about solving high-impact problems by designing tailored, scalable solutions.

EXPERIENCE

Microsoft 2024-present

 Software Engineer Belgrade

- > Developing Guest RDMA enablement on Azure Boost to deliver high-performance networking to customers.
- > Driving design and implementation of low-latency, reliable data paths for RDMA in the Azure cloud.
- > Collaborating with cross-functional teams to integrate and ship Azure networking capabilities at scale.
- > Aligning with stakeholders to meet performance, security, and reliability goals.
- > Implementing, testing, and validating industry-standard RDMA protocols in cloud environments.
- > Acted as DRI for customer escalations, diagnosing technical issues and coordinating resolutions to keep stakeholders satisfied.

Lensed 2022-2024

 Software Engineer San Francisco (remote)

- > Led a technical team to create novel solutions enabling engineer-focused tools for EDA design debugging.
- > Architected and implemented CAD circuit data pipelines that outperform industry solutions.
- > Engineered novel data representation using Rust and Python to seamlessly integrate into toolchains, complying with industry standards.
- > Deployed CI/CD pipelines and delivered production hybrid cloud deployments to customers.

The American University in Cairo 2022-2024

 Graduate teaching assistant Cairo

- > Taught CUDA and parallel programming concepts to students in GPU computing course.
- > Covered advanced concepts including RDMA.
- > Orchestrated deployment of a Docker environment.

Tod-Z 2020-2021

 AI development intern Tallinn (remote)

- > Led AI and machine learning development.
- > Built entire machine learning stack from concept to training.
- > Deployed machine learning models on AWS.

Agile Technology 2020

 Intern Cairo (remote)

- > Built web store using OutSystems.
- > Regularly communicated for quality assurance.

EDUCATION

The American University in Cairo (AUC) 2017-2022

- > BSc in Computer Engineering
- > Graduated with High Honors

SELECTED PROJECTS

ML-based decompiler 2021-2022

   

- > Led a team of 6 to create a machine learning-based decompiler targeting Android.
- > Novel tree graph generator for AST representation.
- > Designed to be a forensic malware tool.

Persistence of vision display 2021

     

- > Created high-speed fan display and embedded system to fulfill microsecond timing constraints.
- > Featured embedded web server for user accessibility and customization.

ML library from scratch 2021

   

- > Created a CUDA-accelerated machine learning library.
- > Implemented training and inference framework for deep neural networks and convolutional neural networks.

Distance sensor 2021

  

- > Built an ultrasonic distance sensor with 1 mm accuracy from scratch.
- > Reverse-engineered market boards to create a more accurate sensor.

RISC-V microarchitecture 2021

  

- > Implemented a 6-stage pipeline RISC-V microarchitecture on an FPGA in Verilog that meets specification requirements.
- > Compiled and ran C programs on FPGA using microarchitecture.

RESEARCH PROJECTS

A Neural Generation Framework for Android De-compilation 2022

- > Completed Bachelor thesis
- > Created a novel method for tree graph generation.
- > Led a team of 4 to create a process that builds application-specific RISC-V-based CPUs.
- > Applied the novel method and transformer translation to Android decompilation.

Application-specific CPU optimizer 2020-2022

- > Worked on a framework to optimize unused hardware in HDL according to an application use case.
- > Led a team of 3 to create an FPGA-focused optimization framework for RISC-V CPUs.

EXTRACURRICULAR

ROV media head 2017-2019

-  Cairo
- > Led a team of 5 to market and document Robotics club's remotely operated vehicle events.
- > Photographed and documented dozens of events and directed 4 marketing videos.

Assistant media head 2017

-  Cairo
- > Lead photographer for AUC newspaper.
- > Published over a dozen photographs covering local events.

NASA camp 2015

-  Atlanta
- > Won "Best Camper" award.

Dive, Eat, Sleep & Repeat 2022

-  Business Monthly Magazine
- > Wrote an article on diving in the Red Sea during shark season.
- > Released in the 2022 August issue of Business Monthly.

PROGRAMMING LANGUAGES

C	Rust	C++
Python	Bash	SQL
LaTeX	Verilog	

COMPETITIONS

HackTrick Hackathon 2022

-  Cairo
- > Built a reinforcement learning model to compete in material pipeline optimization challenges.

ICPC ACM 2019

-  Cairo
- > Entered local ICPC ACM.
- > Was the youngest participant.

Code Geist Hackathon 2019

-  Cairo
- > Created an app and its pitch in 72 hours.
- > Collaborated in a team of 4.
- > Developed a platform to address littering.

MATE ROV 2017-2019

-  Cairo
- > Joined university's Remotely Operated Vehicle team.
- > Contributed to circuit and mechanical design.

Science Fair 2014

-  Cairo
- > Won first place at high school level with a 4-wheeled remote-controlled robot featuring a 3-axis arm.

SKILLS >

LANGUAGES >

HOBBIES

Recreational scuba diver

- > Licensed 2-star CMAS scuba diver.
- > Over 200 dives in the Red Sea since 2009.

Home Lab

- > Built my own liquid-cooled computer.
- > Administered a home lab with file, surveillance, and DNS servers.

3D printing enthusiast

- > Designed my own 3D-printed robot components.

Travel

- > Traveled to most of Europe, the U.S.A., the Middle East, and Africa.